Econometrics 120A: Discussion Section

Midterm 1 Solutions

Anita Sammarini & Linghui Wu

Department of Economics

Covariance and correlation

Covariance and Correlation

Recap

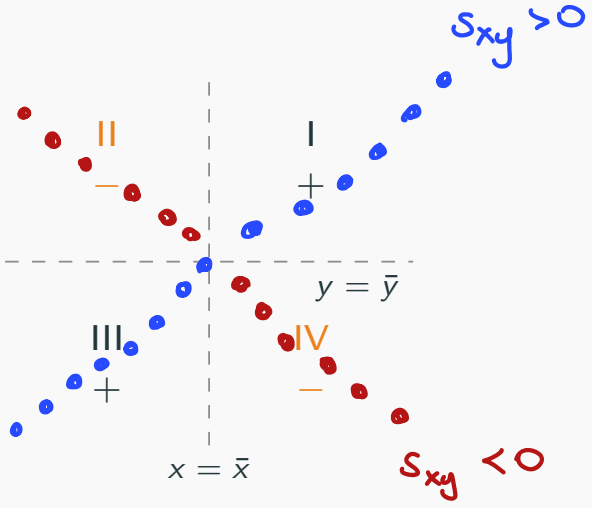
- **Sample covariance:**

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- Each point adds one product $(x_i - \bar{x})(y_i - \bar{y})$. Its sign depends on the quadrant around $(\bar{x}, \bar{y})$.
- **Units:** units of x times units of y.
- **Correlation:**

$$r_{xy} = \frac{s_{xy}}{s_x\, s_y}$$

The units cancel, and r_{xy} is always between -1 and 1.



Sign of $(x_i - \bar{x})(y_i - \bar{y})$ in each quadrant. In I, both factors are $+$; in III, both are $-$; in II and IV, one $+$ and one $-$.

Covariance Questions

Version A, Question 1

Suppose a researcher plots a dataset of $n = 50$ store locations, where x_i is monthly advertising expenditure (in thousands of dollars) and y_i is monthly sales volume (in thousands of units).

(a) (2 pts) If the scatter plot shows data points broadly distributed evenly across all four quadrants defined by the sample means $(\bar{x}, \bar{y})$, what does this imply about the magnitude of the sample covariance s_{xy}?

(b) (2 pts) Mathematically explain how the terms $(x_i - \bar{x})(y_i - \bar{y})$ contribute to the sample covariance sum across the four quadrants relative to $(\bar{x}, \bar{y})$.

Covariance Questions

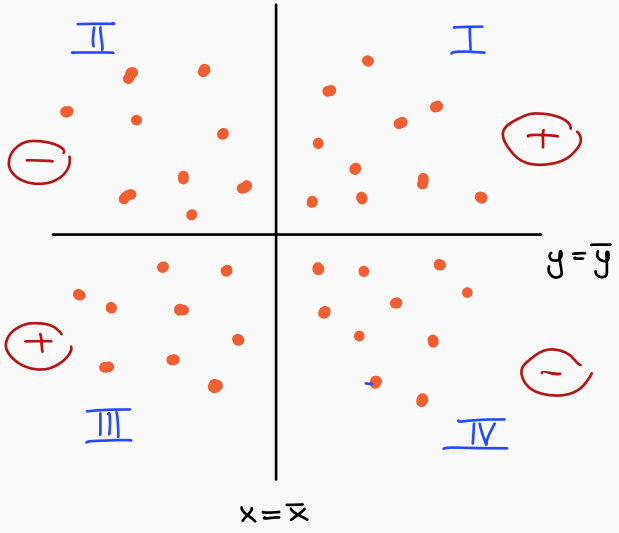
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(a) (2 pts) If the scatter plot shows data points broadly distributed evenly across all four quadrants defined by the sample means $(\bar{x}, \bar{y})$, what does this imply about the magnitude of the sample covariance s_{xy}?

$s_{xy} = 0$

$$s_{xy} = \frac{1}{49} \sum_{i=1}^{50} (x_i - \bar{x})(y_i - \bar{y})$$



Covariance Questions

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Suppose a researcher plots a dataset of $n = 50$ store locations, where x_i is monthly advertising expenditure (in thousands of dollars) and y_i is monthly sales volume (in thousands of units).

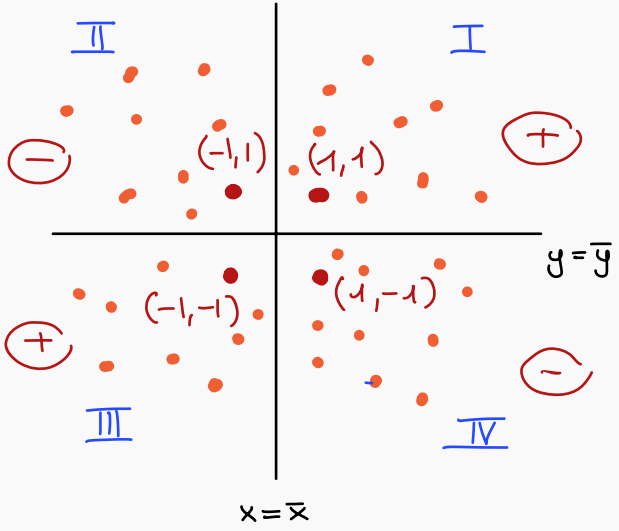
(b) (2 pts) Mathematically explain how the terms $(x_i - \bar{x})(y_i - \bar{y})$ contribute to the sample covariance sum across the four quadrants relative to $(\bar{x}, \bar{y})$.

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} \underbrace{(x_i - \bar{x})}_{\text{horizontal deviation}} \underbrace{(y_i - \bar{y})}_{\text{vertical deviation}}$$

Quadrant	$x_i - \bar{x}$	$y_i - \bar{y}$
I	+	+
II	−	+
III	−	−
IV	+	−

$$s_{xy} = \frac{1}{n-1} \left(\sum_{\text{I, III}} \text{positive terms} + \sum_{\text{II, IV}} \text{negative terms} \right)$$

$$s_{xy} = \frac{1 - 1 + 1 - 1}{3} = 0$$



Covariance and Correlation Questions

Version B, Question 1

(a) (1 pt) Consider two continuous variables measured on a sample of n individuals: height X (measured in meters, m) and weight Y (measured in kilograms, kg). Explain what a positive sample covariance ($s_{xy} > 0$) versus a negative sample covariance ($s_{xy} < 0$) reveals about the directional relationship between height and weight.

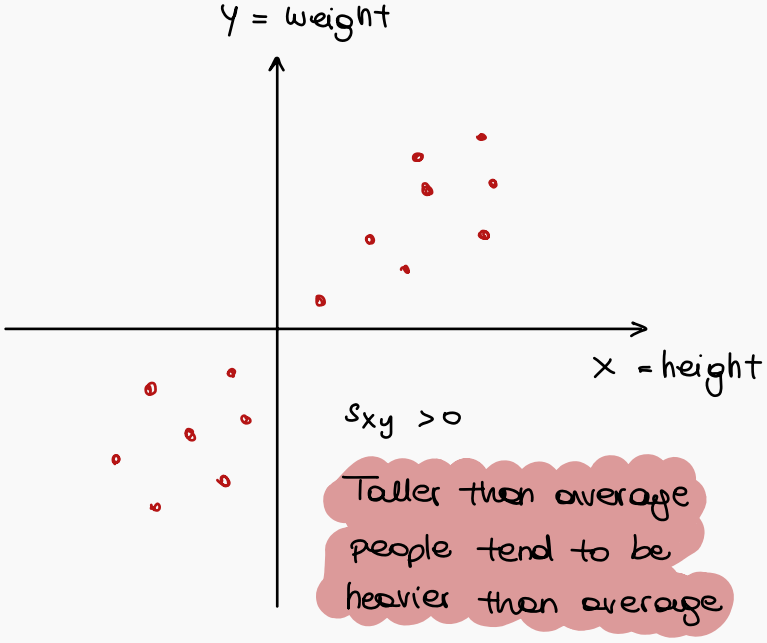
(b) (1 pt) What are the exact units of measurement for the sample covariance s_{xy} in this scenario?

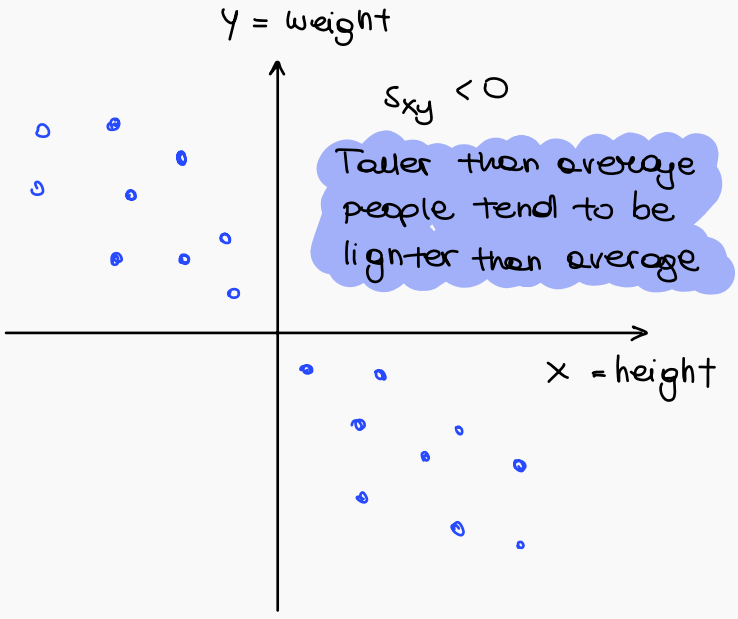
(c) (2 pts) Why is sample covariance alone insufficient for comparing the strength of linear relationships across different pairs of variables (e.g., height vs. weight versus temperature vs. energy consumption)? How does the correlation coefficient (r_{xy}) resolve this limitation?

Covariance and Correlation Questions

Version B, Question 1

(a) (1 pt) Consider two continuous variables measured on a sample of n individuals: height X (measured in meters, m) and weight Y (measured in kilograms, kg). Explain what a positive sample covariance ($s_{xy} > 0$) versus a negative sample covariance ($s_{xy} < 0$) reveals about the directional relationship between height and weight.





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(b) (1 pt) What are the exact units of measurement for the sample covariance s_{xy} in this scenario?

- $s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$ — $(x_i - \bar{x})$: meters; $(y_i - \bar{y})$: kilograms
- Units of s_{xy} = $m \cdot kg$

Covariance and Correlation Questions

Version B, Question 1

(a) (1 pt) Consider two continuous variables measured on a sample of n individuals: height X (measured in meters, m) and weight Y (measured in kilograms, kg). Explain what a positive sample covariance ($s_{xy} > 0$) versus a negative sample covariance ($s_{xy} < 0$) reveals about the directional relationship between height and weight.

(c) (2 pts) Why is sample covariance alone insufficient for comparing the strength of linear relationships across different pairs of variables (e.g., height vs. weight versus temperature vs. energy consumption)? How does the correlation coefficient (r_{xy}) resolve this limitation?

The size of s_{xy} depends on the units !!

$$X_m = 100 \cdot X_{cm} \quad \rightarrow \quad s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (100 x_i - 100 \bar{x})(y_i - \bar{y}) = 100\, s_{x_m, y}$$

Numerical value of the covariance increased by a factor of 100 even though the underlying relationship is unchanged !

$$r_{xy} = \frac{s_{xy}}{s_x \cdot s_y} = \frac{100\, s_{x_m y}}{100\, s_{x_m} \cdot s_y} = r_{x_m, y}$$

$s_{x_{cm}} = 100\, s_{x_m}$

UNITS CANCEL OUT !

Binomial

Binomial Random Variable

Recap: the binomial table

If $S \sim \text{Binomial}(n, p)$, the table gives $P(S = s)$: ❶ block of your n, ❷ column of your p, ❸ row of your s.

$n = 4$

$p = 0.4$

n	s	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
4	0	0.6561	0.4096	0.2401	0.1296	0.0625	0.0256	0.0081	0.0016	0.0001
4	1	0.2916	0.4096	0.4116	0.3456	0.2500	0.1536	0.0756	0.0256	0.0036
4	2	0.0486	0.1536	0.2646	0.3456	0.3750	0.3456	0.2646	0.1536	0.0486
4	3	0.0036	0.0256	0.0756	0.1536	0.2500	0.3456	0.4116	0.4096	0.2916
4	4	0.0001	0.0016	0.0081	**0.0256**	0.0625	0.1296	0.2401	0.4096	0.6561
8	5	0.0004	0.0092	0.0467	0.1239	0.2188	0.2787	0.2541	0.1468	0.0331
8	6	0.0000	0.0011	0.0100	0.0413	0.1094	0.2090	0.2965	0.2936	**0.1488**
8	7	0.0000	0.0001	0.0012	0.0079	0.0313	0.0896	0.1977	0.3355	**0.3826**
8	8	0.0000	0.0000	0.0001	0.0007	0.0039	0.0168	0.0576	0.1678	**0.4305**

$P(S \le 5) \sim 1 - [0.1488 + 0.3826 + 0.4305]$

The table only gives $P(S = s)$. For "at least" or "at most", add rows or use the complement: $P(S \le 5) = 1 - \big[P(S = 6) + P(S = 7) + P(S = 8)\big]$.

$$P(S \ge k) = \sum_{s=k}^{n} P(S = s) \qquad P(S \le k) = 1 - P(S \ge k+1)$$

$$P(S \le k) = \sum_{s=0}^{k} P(S = s)$$

Binomial Questions

Version A, Question 3

A standard medical treatment for a specific skin condition is known to have a 40% recovery rate within one week. A dermatologist tries a newly formatted ointment on a group of 4 independent patients, and all 4 patients successfully recover within the first week. The pharmaceutical representative says this completely confirms the new formula is vastly superior to the 40% standard baseline.

(a) (2 pts) What is the probability that an outcome this successful (or better) might occur purely by chance under the baseline 40% rate?

(b) (2 pts) Based on the probability in (a), do you believe the representative's claim that this proves the new ointment is superior?

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(a) (2 pts) What is the probability that an outcome this successful (or better) might occur purely by chance under the baseline 40% rate?

$S \sim \text{Binomial}(4, 0.4)$

"Successful or better": $P(S \geq 4) = P(S = 4)$

$P(S = 4) = \binom{4}{4}(0.4)^4(0.6)^0 = 0.0256 = 2.56\%$

Assuming the baseline recovery rate is 40% observing four recoveries out of four happens with probability 2.56%.

Binomial Questions

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(b) (2 pts) Based on the probability in (a), do you believe the representative's claim that this proves the new ointment is superior?

2.56% is rare so the result is evidence that the ointment works better than 40%

BUT it DOES NOT PROVE it is vastly superior: 4 patients is a very small sample

Binomial Questions

Version B, Question 3

A tech company asserts that its new automated software testing tool has a 90% success rate in detecting critical bugs. A quality assurance team deploys the tool on 8 independently developed code modules containing known critical bugs. Surprisingly, the tool fails to detect a bug in 3 of these modules (meaning it only successfully detected bugs in 5 modules). The developers maintain that the tool's 90% claim is still valid.

(a) (2 pts) What is the probability that such an outcome (failing to detect 3 or more bugs) might occur?

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(a) (2 pts) What is the probability that such an outcome (failing to detect 3 or more bugs) might occur?

$n = 8$

$p = 0.9$

S = number of successfully detected bugs

3 failures $\Leftrightarrow$ $S = 5$

$$P(S \leq 5) = 1 - P(S > 5) =$$
$$= 1 - P(S \geq 6) =$$
$$= 1 - [P(S=6) + P(S=7) + P(S=8)] =$$
$$= 1 - [0.1488 + 0.3826 + 0.4305] =$$
$$= 1 - 0.9619 = 0.0381 = 3.81\%$$

Version B, Question 3

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(b) (2 pts) Based on the probability in (a), do you believe that the company's claim is still correct?

Not impossible but DOUBTFUL!

A result this bad should happen fewer than 3.8% of the time if the 90% claim were true.

Normal

Normal Random Variable

Recap: two kinds of normal questions

$X \sim N(\mu, \sigma^2)$

$Z = (X - \mu)/\sigma \sim N(0, 1)$

Table: $P(Z > z)$, only for $z \geq 0$

Given the probability find X!

1. Find x from a probability $P(X > x)$
   - ⇨ "Top 8%" $\rightarrow P(X > x) = 0.08$
   - ⇨ Find 0.08 inside the table, read the closest z
   - ⇨ Then $x = \mu + z\,\sigma$

Given x find the probability

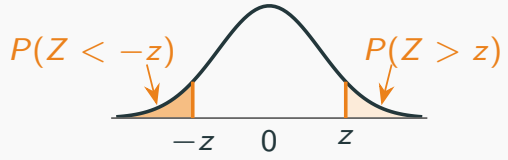
2. Find $P(X < x)$ given x
   - ⇨ If $z = (x - \mu)/\sigma$ is positive: $1 - P(Z > z)$
   - ⇨ If $z = (x - \mu)/\sigma$ is negative, flip it: $P(Z < -z) = P(Z > z)$

z	0.00	0.01	0.02	0.03	0.04
1.0	0.1587	0.1562	0.1539	**0.1515**	**0.1492**
1.1	0.1357	0.1335	0.1314	0.1292	0.1271
1.2	0.1151	0.1131	0.1112	0.1093	0.1075
1.3	0.0968	0.0951	0.0934	0.0918	0.0901
→ 1.4	**0.0808**	**0.0793**	0.0778	0.0764	0.0749
2.0	**0.0228**	0.0222	0.0217	0.0212	0.0207
2.1	0.0179	0.0174	0.0170	0.0166	0.0162
2.2	0.0139	0.0136	0.0132	0.0129	0.0125
2.3	0.0107	0.0104	0.0102	**0.0099**	0.0096

Row: first decimal of z

Column: second decimal of z

2. Negative z: the two tails have the same area



Normal Questions

Version A, Question 4

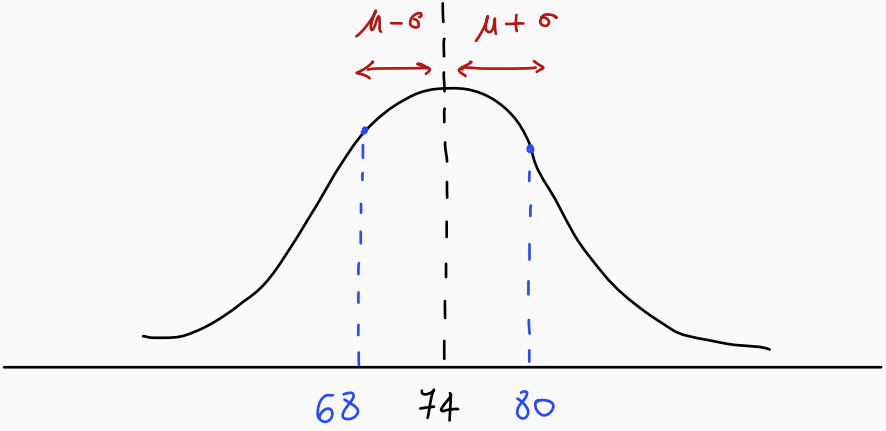
$\mu = 74$
$\sigma = 6$

$X \sim N(74, 6)$

Transit times for a specific cross-country shipping route average 74 hours. Assume that these transit times are normally distributed with a standard deviation of 6 hours.

(a) (2 pts) What is the transit time of a shipment at the top 8th percentile of slowest shipments? (i.e., if 8% of shipments took longer than yours, what was your transit time?)

(b) (2 pts) What is the proportion of shipments that arrive in less than 60 hours?



Version A, Question 4

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Given probability, find x (1)

$P(X > x) = 0.08$ or $P(X \leq x) = 1 - 0.08 = 0.92$

standardize!

$$P(X > x) = P\left(\frac{X-\mu}{\sigma} > \frac{x-\mu}{\sigma}\right) = P\left(z > \frac{x-\mu}{6}\right) = 0.8$$

$$P\left(z > \frac{x-74}{6}\right) = 0.08$$

From the table: $P(z > 1.40) = 0.0808$, $P(z > 1.41) = 0.0793$ $\rightarrow$ $z \approx 1.405$

$$z = \frac{x-\mu}{6} \Rightarrow x = \mu + 6z$$

$$= 74 + 6(1.405) = \underline{82.43 \text{ hours}}$$

Version A, Question 4

Transit times for a specific cross-country shipping route average 74 hours. Assume that these transit times are normally distributed with a standard deviation of 6 hours.

(b) (2 pts) What is the proportion of shipments that arrive in less than 60 hours?

Given X find probability (2)

$$P(X < 60) = P\left(Z < \frac{60-\mu}{\sigma}\right) =$$

$$= P\left(Z < \frac{60-74}{6}\right) =$$

$$= P(Z < -2.33) =$$

normal is symmetric around zero

$$= P(Z > 2.33) =$$

$$\approx 0.0099 = 0.99\%$$

Version B, Question 4

The lifespan of a standard delivery van's brake pads averages 32,000 miles. Assume that brake pad lifespans are normally distributed with a standard deviation of 3,500 miles.

$X \sim N(32000, 3500)$

(a) (2 pts) What is the lifespan of brake pads at the top 15th percentile? (i.e., if 15% lasted longer than yours, what would their lifespan be?)

(b) (2 pts) What is the proportion of vans whose brake pads wear out in less than 25,000 miles?

Version B, Question 4

The lifespan of a standard delivery van's brake pads averages 32,000 miles. Assume that brake pad lifespans are normally distributed with a standard deviation of 3,500 miles.

(a) (2 pts) What is the lifespan of brake pads at the top 15th percentile? (i.e., if 15% lasted longer than yours, what would their lifespan be?)

Given probability, find x (1)

$$P(X > x) = 0.15 \quad \text{or} \quad P(X \leq x) = 0.85$$

$$P\left(z > \frac{x - 32000}{3500}\right) = 0.15$$

From the table $P(z > 1.03) = 0.1515$, $P(z > 1.04) = 0.1492$ $\Rightarrow z \approx 1.0366$

$$x = \mu + 6z = 32000 + 3500(1.0366) = 35\,628 \text{ miles}$$

Approximately 15% of brake pads last longer than 35 628 miles

Version B, Question 4

The lifespan of a standard delivery van's brake pads averages 32,000 miles. Assume that brake pad lifespans are normally distributed with a standard deviation of 3,500 miles.

(b) (2 pts) What is the proportion of vans whose brake pads wear out in less than 25,000 miles?

Given X find probability (2)

$$P(X < 25000) = P\left(Z < \frac{25000 - 32000}{3500}\right) = P(Z < -2) =$$

$$= P(Z > 2) =$$

$$= 0.0228 =$$

$$= 2.28\%$$

Continuous densities

Continuous Random Variable

Recap: when is $p(x)$ a valid density?

A function $p(x)$ is a valid density only if **both** rules hold:

1. **Never negative:** $p(x) \geq 0$ for every x.
2. **Total area is 1:**

$$\int_{-\infty}^{\infty} p(x)\,dx = 1$$

Checking only the area is not enough. Probabilities are areas under $p(x)$, and a probability can never be negative.

Reminder: $\int x^k\,dx = x^{k+1}/(k+1)$.

$$P(a < X < b) = \int_a^b p(x)\,dx$$

Continuous Random Variable Questions

New question

Suppose that we have a function $p(x) = x - \frac{1}{2}$ where $0 < x < 2$.

(a) Show that the area under $p(x)$ is equal to 1.

(b) Is $p(x)$ a valid density? Explain.

New question

Suppose that we have a function $p(x) = x - \frac{1}{2}$ where $0 < x < 2$.

(a) Show that the area under $p(x)$ is equal to 1.

$$\int_0^2 p(x)\,dx = \int_0^2 \left(x - \frac{1}{2}\right)dx = \left[\frac{x^2}{2} - \frac{1}{2}x\right]_0^2 =$$

$$= \left[\frac{4}{2} - \frac{2}{2} - 0\right] = (2-1) = \underline{\underline{1}}$$

Continuous Random Variable Questions

New question

Suppose that we have a function $p(x) = x - \frac{1}{2}$ where $0 < x < 2$.

(b) Is $p(x)$ a valid density? Explain.

$p(x) \geq 0 \quad \forall x$???

Let's check → $x - \frac{1}{2} \geq 0 \iff x \geq \frac{1}{2}$ → NOT $\forall x$, NOT A VALID DENSITY

Why it matters:

$$\int_0^{1/2} p(x)\,dx = \left[\frac{x^2}{2} - \frac{1}{2}x \right]_0^{1/2} = \frac{1}{8} - \frac{1}{4} = -\frac{1}{8} < 0 \Rightarrow P(0 < X < 1/2) < 0$$

IMPOSSIBLE!

Answer key (TA only)

Answer key

Covariance: Version A Q1 and Version B Q1

A1(a) The products are positive in quadrants I and III and negative in quadrants II and IV. With points spread evenly over all four, they cancel, so s_{xy} is close to 0: no linear relationship between advertising and sales in this sample.

A1(b) $s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$. Quadrant I ($x_i > \bar{x}$, $y_i > \bar{y}$): $(+)(+)$, so a positive term. Quadrant III ($x_i < \bar{x}$, $y_i < \bar{y}$): $(-)(-)$, so a positive term. Quadrant II ($x_i < \bar{x}$, $y_i > \bar{y}$): $(-)(+)$, so a negative term. Quadrant IV ($x_i > \bar{x}$, $y_i < \bar{y}$): $(+)(-)$, so a negative term. An even spread means the positive and negative terms roughly cancel, and the sum is near 0.

B1(a) $s_{xy} > 0$: people taller than average ($x_i > \bar{x}$) tend to also be heavier than average ($y_i > \bar{y}$). $s_{xy} < 0$: people taller than average tend to be lighter than average ($y_i < \bar{y}$).

B1(b) Each term is (meters) $\times$ (kilograms), so s_{xy} is in m $\cdot$ kg (meter-kilograms).

B1(c) The size of s_{xy} depends on the units: measure height in centimeters instead of meters and s_{xy} is 100 times bigger, with the same data. So covariances of different pairs, in different units, cannot be compared. The correlation $r_{xy} = s_{xy}/(s_x s_y)$ divides by both standard deviations, so the units cancel. It has no units and is always between -1 and 1, so it measures only the strength of the linear relationship.

Answer key

Binomial: Version A Q3 and Version B Q3

A3(a) Let S = number of the 4 patients who recover. If the 40% rate holds and patients are independent, $S \sim \text{Binomial}(4, 0.4)$. "This successful or better" is $S \geq 4$, which here is only $S = 4$: $P(S = 4) = 0.4^4 = 0.0256$ (table: $n = 4$, $s = 4$, $p = 0.4$). This happens about 2.6% of the time by chance.

A3(b) 2.56% is rare, so the result is evidence that the ointment works better than 40%. But it does not "prove" it is "vastly superior": 4 patients is a very small sample, and 2.56% is not 0. A larger trial is needed.

B3(a) Let S = number of bugs detected out of 8. If the claim holds and the modules are independent, $S \sim \text{Binomial}(8, 0.9)$. Failing on 3 or more means detecting 5 or fewer, $S \leq 5$. Use the complement:

$$\begin{aligned} P(S \leq 5) &= 1 - \big[P(S = 6) + P(S = 7) + P(S = 8)\big] \\ &= 1 - (0.1488 + 0.3826 + 0.4305) = 1 - 0.9619 = 0.0381 \end{aligned}$$

About 3.8% of the time.

B3(b) If the 90% claim were true, a result this bad or worse would happen fewer than 1 time in 25. It is not impossible, but the claim is doubtful: the true detection rate is likely below 90%.

Answer key

Normal: Version A Q4 and Version B Q4

A4(a) $X \sim N(74, 6^2)$. We need x with $P(X > x) = 0.08$, so $P\left(Z > \frac{x-74}{6}\right) = 0.08$. Table: $P(Z > 1.40) = 0.0808$ and $P(Z > 1.41) = 0.0793$, so $z \approx 1.405$ (1.41 is fine).
$x = 74 + 6 \times 1.405 = 82.43$ hours.

A4(b) $P(X < 60) = P\left(Z < \frac{60-74}{6}\right) = P(Z < -2.33) = P(Z > 2.33) = 0.0099$. About 1% of shipments arrive in less than 60 hours.

B4(a) $X \sim N(32000, 3500^2)$. We need x with $P(X > x) = 0.15$, so $P\left(Z > \frac{x-32000}{3500}\right) = 0.15$. Table: $P(Z > 1.03) = 0.1515$ and $P(Z > 1.04) = 0.1492$, so $z \approx 1.0366$ (1.04 is fine).
$x = 32000 + 3500 \times 1.0366 = 35{,}628$ miles (35,640 with $z = 1.04$).

B4(b) $P(X < 25000) = P\left(Z < \frac{25000-32000}{3500}\right) = P(Z < -2.00) = P(Z > 2.00) = 0.0228$. About 2.3% of vans.

Answer key

New question: $p(x) = x - \frac{1}{2}$ on $0 < x < 2$

(a)

$$\int_0^2 \left(x - \frac{1}{2}\right) dx = \left[\frac{x^2}{2} - \frac{x}{2}\right]_0^2 = \left(\frac{4}{2} - \frac{2}{2}\right) - 0 = 2 - 1 = 1$$

(b) No. The area is 1, but $p(x)$ is negative for every x between 0 and $\frac{1}{2}$. For example, $p(0.25) = 0.25 - 0.5 = -0.25 < 0$. A valid density must be ≥ 0 everywhere, so $p(x)$ fails the first rule.

Why it matters: the area from 0 to $\frac{1}{2}$ is $\int_0^{1/2}(x - \frac{1}{2})\,dx = \frac{1}{8} - \frac{1}{4} = -\frac{1}{8}$. So "$P(0 < X < \frac{1}{2})$" would be negative, which no probability can be. The extra area to the right of $\frac{1}{2}$ hides this when we only check the total.

Compare with Version B Q5: $p(x) = \frac{3}{8}x^2 \geq 0$ on $0 < x < 2$ and its area is 1, so it passes both rules.